

Solar Activity & Space Weather Monitoring System

Objective

Design and implement a reproducible monitoring and analytics system for solar activity and space weather indicators, with the aim of understanding temporal variability, coupling between solar and geospace phenomena, and the challenges of working with multi-source physical time-series data.

Core Tasks

- Identify and ingest **publicly available solar and space weather datasets** (e.g., sunspot numbers, solar flare records, geomagnetic indices such as Kp or Dst) from authoritative sources.
- Harmonise datasets with differing temporal resolutions, units, and reporting conventions.
- Design a modular data-cleaning pipeline to handle missing observations, reporting gaps, and known artefacts.
- Aggregate and align indicators across multiple temporal scales (hourly, daily, monthly).
- Analyse relationships between solar activity indicators and space weather responses using time-series comparison and correlation analysis.
- Build an interactive dashboard that visualises:
 - Individual solar and geomagnetic indicators over time
 - Cross-indicator comparisons and lagged relationships
 - Periods of heightened or anomalous activity

Intensification Requirements

- Implement and compare **at least two smoothing or baseline estimation strategies** (e.g., rolling statistics vs cycle-based baselines) and assess their impact on interpretation.
- Identify and characterise **episodes of extreme solar or geomagnetic activity**, situating them in their broader temporal context.
- Analyse **lead-lag relationships** between solar drivers (e.g., flares, sunspots) and geomagnetic responses.
- Explicitly discuss data limitations, including measurement uncertainty, index construction, and observational coverage.
- Include at least one **iterative refinement** where aggregation choices, indicator selection, or visual encodings are revised based on exploratory findings.

Expected Outcome

A well-documented solar activity and space weather monitoring dashboard that demonstrates principled time-series data handling, multi-source integration, and informed interpretation of coupled physical systems.